

Modulating the Stability of Collagen Triple Helices by Terminal Charged Residues

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Cite This: *J. Phys. Chem. B* 2021, 125, 7351–7358



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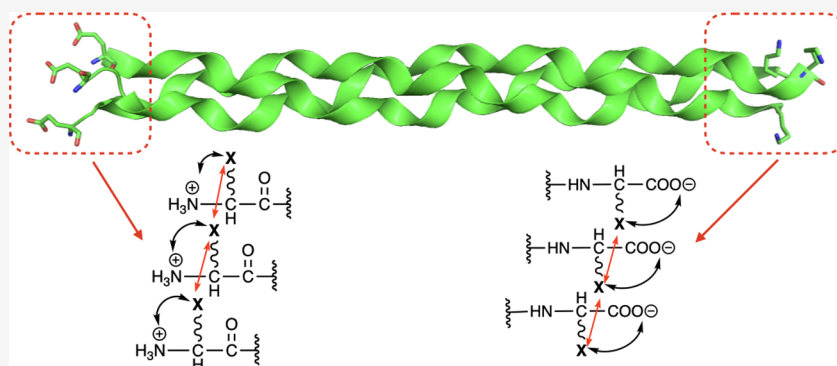
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ABSTRACT: Cationic or anionic residues are frequently located at the termini of proteins because their charged side chain can form electrostatic interactions with a terminal carboxylate or ammonium group to stabilize the structure under physiological conditions. Here, we used collagen-mimetic peptides (CMPs) to examine how the terminal charge–charge interactions affect the collagen triple helix stability. We designed a series of CMPs with either a Lys or Glu incorporated into the terminus and measured their pH-dependent stability. The results showed that the **terminal electrostatic attractions stabilized the triple helix**, while the terminal electrostatic repulsions destabilized the trimer. The data also revealed that the repulsions imposed a greater effect than did the attractions on the triple helix. An amino acid with a shorter side chain, such as aspartate and ornithine, was also installed to investigate the length effect on electrostatic interactions, which was found to be insignificant. Meanwhile, simultaneously incorporating cationic and anionic residues into the termini showed slight additive stabilization effects but pronounced additive destabilization consequences. We have demonstrated that the collagen triple helix stability can be modulated by introducing a cationic or anionic residue into the terminus of a peptide, giving useful information for the design of collagen-associated materials.

INTRODUCTION

As the most abundant protein in mammals, collagen has been widely studied with collagen-mimetic peptide (CMP) models in the past decades.^{1–3} It is a triple helix composed of three polyproline II-like strands. Each strand contains numerous X-Y-Gly repeats, in which X is often proline (Pro), while Y is mostly (2S,4R)-hydroxyproline (Hyp, O).⁴ The use of short CMPs to study the collagen structure has revealed various factors and forces involved in stabilizing the triple helix and provided many insights into the interactions within the trimeric fold, making CMP a great model to mimic natural collagen.^{1–3} For instance, the replacement of Pro with Hyp in the Y position significantly stabilizes the trimer. This stabilization has been attributed to the C'-exo puckering of Hyp, stereoelectronic effects, and steric interstrand repulsions.^{3,5–10} Because Gly is a sterically nondemanding residue within the triple helix and is a hydrogen bond donor to the carbonyl group on the adjacent strand, mutating Gly to other residues significantly destabilizes the triple helix and even leads to an unfolded structure.^{11,12} Thus, the Gly residue required in

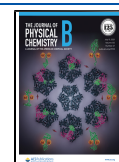
every third position plays a crucial role in maintaining a stable triple helix.

Besides the mutation studies, a recent work by Wennemers and co-workers demonstrated that the charged ($-\text{COO}^-$ and $-\text{NH}_3^+$) and neutral functional groups ($\text{CH}_3\text{CO}-$ and $-\text{CONH}_2$) at the N- and C-termini significantly affect the stability of the triple helix.¹³ This study further suggests that the charge–charge interactions between three collagen peptides contribute to the folding stability and that neutralizing the terminal charge can stabilize the triple helical conformation. For an α -helix, introducing a positively charged residue into its negative C-terminus is much more favorable than

Received: February 23, 2021

Revised: June 17, 2021

Published: July 2, 2021



putting such a residue into its positive N-terminus.^{14,15} An N-cap Asp residue was also previously shown to stabilize a small α/β protein because of the electrostatic interactions between its side chain carboxylate and the terminal ammonium.¹⁶ In addition, the charges at the peptide termini were shown to have a substantial effect on polyproline helices.¹⁷ Recently, we also used a series of polyproline peptides with a cationic or anionic residue installed at the terminus to show that a C-terminal Lys or Arg can stabilize the polyproline II helix and prohibit the transformation into a polyproline I helix.¹⁸ This again indicates the importance of the terminal electrostatic interaction on a peptide structure and also reveals that such a force may be used to tune the folding stability of a protein or peptide.

To investigate the influence of a terminal charged residue on the stability of a collagen triple helix, herein we designed and prepared CMPs with the residue with a **cationic side chain** (Lys) or **an anionic side chain** (Glu) incorporated into the N- or C-terminus. As shown in Figure 1, within a triple helix, the

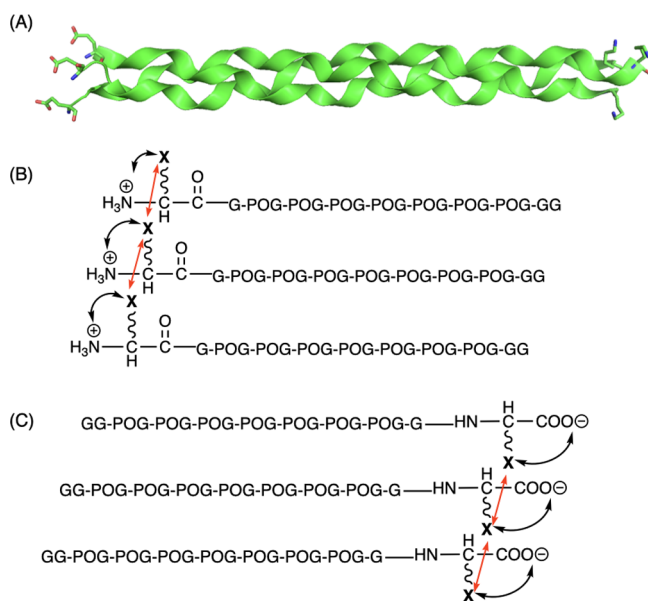


Figure 1. (A) Diagram of a collagen triple helix. The terminal residues were replaced by anionic or cationic residues shown in sticks. The diagram was generated by PyMOL 2.4 using the crystal structure of a collagen model (PDB: 1CAG). (B) Charge–charge interactions at the N-terminus. (C) Charge–charge interactions at the C-terminus. The black arrow indicates the interaction between the charged side chain (X) and the terminal group, while the red arrow represents the interaction between charged side chains.

charged side chain of the introduced residue will interact with the terminal functional group and the same charged side chain on the adjacent strand. According to the structure of collagen, as shown in Figure 1A, the interior of a triple helix is relatively tightly packed, so the side chains on C $^{\alpha}$ are essentially exposed to the solvent and oriented outside the triple helix. Previous studies have shown that the interchain cationic–anionic and interchain cation– π interactions can induce the formation of collagen heterotrimers,^{19–24} and interstrand covalent bonds between cationic and anionic residue side chains can form upon mixing the CMPs to assist self-assembly.^{25,26} These findings further indicate that the installed terminal residue will have its side chain oriented to the exterior of the triple helix, allowing us to examine the impacts of short-range side chain–

terminus and long-range side-chain–side-chain interactions on the triple helix via our designed CMPs. Previous works had used the cationic and anionic residues in various side-chain lengths to show the length effects on the ion–pair interactions and the stability of short α -helices, β -sheets, and collagen triple helices.^{27–33} Therefore, we replaced lysine with ornithine and glutamic acid with aspartic acid to explore the side-chain length effects on the terminal charge–charge interactions within a collagen triple helix. Finally, in the present work, we also incorporated both cationic and anionic residues into the termini of a CMP to examine whether the consequences were additive.

MATERIALS AND METHODS

Synthesis of amino acid-attached resin: Fmoc-Asp(OtBu)-OH, Fmoc-Glu(OtBu)-OH, Fmoc-Gly-OH, Fmoc-Lys(Boc)-OH, and Fmoc-Orn(Boc)-OH were attached to 2-chlorotriptyl resin.

A solution of Fmoc-protected amino acid (1 mmol) and *N,N*-diisopropylethylamine (0.52 mL, 2.88 mmol) in dry CH₂Cl₂ (~5 mL) was added to a solution in which 1 mmol 2-chlorotriptyl resin (297 mg, loading 1.5 mmol/g) was swollen in dry CH₂Cl₂ (~7 mL) under N₂(g). The mixture was stirred under N₂(g) at room temperature for approximately 2 h, and then CH₃OH was added to cap any remaining active sites on the resin. The product was isolated by filtration, washed with dry CH₂Cl₂ (~50 mL), and dried under reduced pressure. Loading capacity was determined by the absorbance of the residual solution from resin Fmoc cleavage at 300 nm using UV–vis spectroscopy and an absorption coefficient of 6566 M^{−1} cm^{−1} for Fmoc. According to the following relationship

$$\text{substitution level (mmol/g)} = (3.05 \times A_{300})/m$$

$$= \text{mass of resin (mg)}, A_{300} = \text{absorbance at 300 nm}$$

the determined loading capacity was 0.742 mmol/g for the Fmoc-Asp(OtBu)-resin, 0.823 mmol/g for the Fmoc-Glu(OtBu)-resin, 0.618 mmol/g for the Fmoc-Gly-resin, 0.509 mmol/g for the Fmoc-Lys(Boc)-resin, and 0.764 mmol/g for the Fmoc-Orn(Boc)-resin, respectively.

Peptide Preparation. Chemical reagents and fluorenylmethoxycarbonyl (Fmoc)-protected amino acids were purchased from Sigma-Aldrich, J. T. Baker, Advanced Chemtech, and CreoSalus. All peptides were prepared on a 0.1 mmol scale by solid-phase methods and Fmoc chemistry protocols using Fmoc-protected amino acids, and HOBt/HBTU-mediated coupling reactions on a PS3 peptide synthesizer (Protein Technologies Inc.). The use of amino acid-preloaded 2-chlorotriptyl resin generated a free C-terminus, and Rink amide resin generated an amidated C-terminus upon cleavage from the resin with a solution of 95% trifluoroacetic acid (TFA)/2.5% triisopropylsilane/2.5% H₂O (v/v). Each peptide had a free N-terminus except for acKT7 and acET7, whose N-termini were acetylated before cleavage from the resin. All peptides were purified by reverse-phase high-performance liquid chromatography (HPLC) with a Vydac or Dr. Maisch semipreparative C18 column. H₂O/acetonitrile gradients with 0.1% (w/v) TFA as the counterion were used for the eluting solvent system. All the peptides were >90% pure according to analytic HPLC analysis. Identities of all the peptides were confirmed by matrix-assisted laser desorption/ionization time-of-flight mass spectrometry (Autoflex III Smartbeam LRF200-CID, Bruker Daltonics). The calculated

and measured molecular weights of these peptides are shown in Table S1 of Supporting Information.

Circular Dichroism Spectroscopy. Circular dichroism (CD) experiments were conducted with an Aviv model 410 CD spectrometer. Three buffered solutions with different pH values (pH 3.0, 7.0, and 11.0) as well as a pH 7.0 buffered solution containing 2 M NaCl were used to dissolve the peptides. The pH 7.0 and 11.0 buffered solutions contained 20 mM sodium phosphate, while the pH 3.0 solution contained 20 mM acetic acid. A working peptide concentration of 200 μ M was used for CD measurements. CD spectra were collected using a 1 mm pathlength quartz cuvette. Prior to measurements, the peptide solutions were heated at 80 °C for 20 min, cooled to room temperature, and held for 15 min before incubation at 4 °C for more than 24 h. Thermal denaturation curves were obtained by recording ellipticity at 225 or 226 nm in the temperature range of 4–90 °C with an average heating rate of 0.16 °C/min and an interval of 1 °C. The values of T_m were determined by fitting the unfolding curves into a simple two-state model.³⁴ The fitting method is described in the Supporting Information.

RESULTS AND DISCUSSION

Peptide Design. We used a 25-residue CMP, Gly-Gly-(Pro-Hyp-Gly)₇-Gly-Gly, as the parent peptide for reference (T7 in Table 1). The molecular structure of the peptide model

Table 1. Sequences of Collagen-Mimetic Peptides Used in This Study

peptide	sequence ^a
T7	H-GG(POG) ₇ GG-OH
KT7	H-KG(POG) ₇ GG-OH
ET7	H-EG(POG) ₇ GG-OH
acKT7	CH ₃ CO-KG(POG) ₇ GG-OH
acET7	CH ₃ CO-EG(POG) ₇ GG-OH
T7K	H-GG(POG) ₇ GK-OH
T7E	H-GG(POG) ₇ GE-OH
T7Kam	H-GG(POG) ₇ GK-NH ₂
T7Eam	H-GG(POG) ₇ GE-NH ₂
ET7K	H-EG(POG) ₇ GK-OH
KT7E	H-KG(POG) ₇ GE-OH
OrnT7	H-OrnG(POG) ₇ GG-OH
DT7	H-DG(POG) ₇ GG-OH
T7Orn	H-GG(POG) ₇ GOrn-OH
T7D	H-GG(POG) ₇ GD-OH

^aO is (2S,4R)-hydroxyproline and Orn is ornithine. H indicates a free N-terminus and OH indicates a free C-terminus. NH₂ indicates an amidated C-terminus, while CH₃CO indicates an acetylated N-terminus.

is shown in Figure 1. The extra glycine residues were added at the termini as a spacer to avoid the unfavorable steric strain between the POG triplet and the side chain of the installed residue. We realized that the extra Gly-Gly sequence could break the POG repeat and destabilize the triple helix. However, the T_m value of the parent peptide (40.7 °C in Table 2) is almost identical to that of the (POG)₇ we previously studied under a similar condition (42.0 °C),³⁵ indicating that the extra Gly-Gly sequence does not induce considerable destabilization effects. This would make our peptide design appropriate. To investigate the impacts of the terminal charged side chains on the stability of a collagen triple helix, we first designed a

Table 2. Melting Temperatures (T_m) of the Triple Helices Formed by the Collagen-Mimetic Peptides at Different pH Values^a

peptide	pH 3	pH 7	pH 11	pH 7/2.0 M NaCl ^b
T7	38.7 ± 0.1	40.7 ± 0.3	42.5 ± 0.2	41.2
N-terminal charged residue installation				
KT7	30.7 ± 0.4	38.4 ± 0.1	41.4 ± 0.5	40.2
ET7	40.0 ± 0.9	40.7 ± 0.8	40.1 ± 0.9	40.1
acKT7	42.0 ± 0.2	41.6 ± 0.1	43.1 ± 0.4	
acET7	44.4 ± 0.3	40.8 ± 0.1	41.0 ± 0.4	
OrnT7	31.0 ± 0.2	38.1 ± 0.1	40.1 ± 0.4	38.3
DT7	40.5 ± 0.1	41.4 ± 0.1	40.8 ± 0.1	41.2
C-terminal charged residue installation				
T7K	39.0 ± 0.1	42.0 ± 0.1	43.5 ± 0.1	40.9
T7E	39.5 ± 0.2	38.7 ± 0.1	39.7 ± 0.6	40.9
T7Kam	36.5 ± 0.2	39.7 ± 0.4	42.6 ± 0.6	
T7Eam	38.6 ± 0.1	40.1 ± 0.2	41.7 ± 0.1	
T7Orn	37.0 ± 0.3	39.4 ± 0.6	40.8 ± 0.1	39.6
T7D	39.0 ± 0.2	35.8 ± 0.3	37.9 ± 0.1	39.9
N, C-terminal charged residue installation				
ET7K		41.6 ± 0.3		
KT7E		34.2 ± 0.4		

^aThe values after ± indicate the deviations from duplicate or triplicate measurements for the T_m values under different pH conditions. ^bThe T_m values in the presence of 2 M NaCl are from the regression fitting of the individual thermal unfolding curves. The fitting errors are less than ±0.2 °C.

series of CMPs in which either a lysine or a glutamate residue was incorporated into the terminus. Among the designed CMPs, those with an acetylated N-terminus or an amidated C-terminus were used to examine the contributions of the interactions between the terminal functional groups (ammonium or carboxylate) as well as the interstrand-charged side-chain interactions to the triple helix stability. In addition, we also replaced lysine with ornithine and glutamate with aspartate to study how the length of a charged side chain affects the terminal charge–charge interactions. Furthermore, two more CMPs with both Lys and Glu incorporated were prepared to study if the influences of terminal charge–charge interactions on collagen were additive. The designed peptides and their sequences are listed in Table 1. All the peptides were chemically synthesized using solid-phase methods and Fmoc-based chemistry.

Effects of the N-Terminal Charged Residues on the Triple Helix Stability. To study how the N-terminal-charged residues affected the triple helix stability, we prepared four CMPs with either a Lys or a Glu incorporated into the N-terminus: KT7, ET7, acKT7, and acET7. Of these four peptides, two CMPs (KT7 and ET7) carried a free N-terminal amino group, while the other two (acKT7 and acET7) had an acetylated N-terminus. After the preparation of the peptides, the CMPs were characterized by CD. According to the far-UV CD spectra at pH 7 (Figure S1 of Supporting Information), all the peptides had a local maximum between 220 and 230 nm, showing the formation of polyproline II (PPII) structures in solution. From the thermal unfolding measurements monitored by CD at different pH values, these CMPs all exhibited a sigmoidal denaturation curve (Figures S2 and S3 of Supporting Information), indicating that they formed triple helices under these conditions. The N-terminal amino group and the ϵ -NH₂ of Lys would be protonated at physiological and acidic pH,

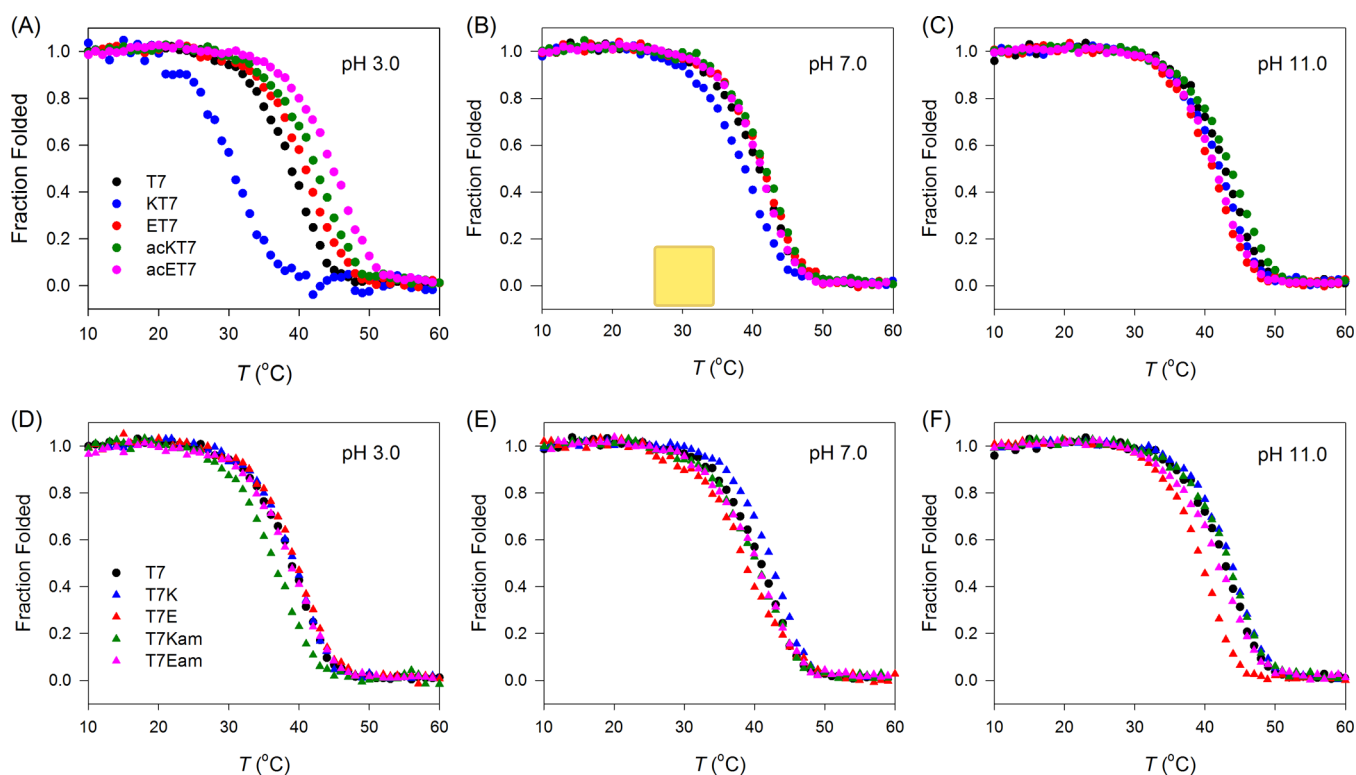


Figure 2. Thermal unfolding curves of the triple helices derived from CMPs at different pH values. The curves are normalized to the folded fraction.

while the γ -COOH of Glu would be deprotonated at physiological and basic pH. Thus, pH-dependent measurements allowed us to examine how the charge repulsions between the positive ammoniums and the charge attractions between the positive ammoniums and the negative carboxylates affected the triple helix.

As shown in Figure 2A–C and Table 2, KT7 has T_m values of 30.7 °C at pH 3, 38.4 °C at pH 7, and 41.4 °C at pH 11, showing that the repulsions between the N-terminal ammonium and the ϵ -NH $_3^+$ of Lys significantly destabilized the triple helix. At pH 3 and pH 7, the triple helix derived from KT7 was less stable than that derived from T7, also indicating that the positive charge–charge repulsions at the N-terminus imposed a destabilization effect on the triple helix. As shown in Figure 3A, the more than 10 °C difference in T_m values between pH 3 and pH 11 indicates that the neutral amino groups diminished charge repulsions and destabilization at pH 11. Moreover, as shown in Table 1, the T_m value of 40.2 °C in the presence of 2.0 M NaCl is close to that at pH 11, supporting the existence of N-terminal charge repulsions that could be reduced by adding concentrated salts. We further used the acetylated N-terminus CMP (acKT7) to differentiate the influence of the ϵ -NH $_3^+$ of Lys and the N-terminal α -NH $_3^+$ on the triple helix. Compared to KT7, the T_m value of acKT7 exhibited an increase at each pH measured: 11.3 °C at pH 3, 3.2 °C at pH 7, and 1.7 °C at pH 11. In particular, the stability difference between KT7 and acKT7 was relatively prominent under an acidic condition (Figure 3A). These results showed that the destabilization effect on the triple helix was much greater when both the N-terminus and the side chain of Lys carried a positive charge, further demonstrating that the multiple positive charges at the N-terminus would have a

detrimental effect on the stability of triple helices formed from short CMPs.

In contrast, ET7 had very close melting temperatures at the three different pH values, indicating that the side chain γ -COO $^-$ of Glu did not have a considerable effect on the triple helix from the N-terminus. The pH-independent stability of ET7 suggested that there were no significant electrostatic interactions between the N-terminal α -NH $_3^+$ and the γ -COO $^-$ of Glu generated to stabilize the triple helix, which was also supported by the result that, at pH 7, ET7 had an almost identical T_m value to that of T7 and adding concentrated NaCl only slightly changed its T_m value. Additionally, by comparing the melting temperatures of ET7 and acET7 at pH 3 and pH 11, we found that removing the N-terminal positively charged ammonium made a positive contribution to the triple helix stability, while the negatively charged side chain of Glu only slightly affected the folding stability. The results that acKT7 and acET7 formed more stable triple helices than did their uncapped counterparts (KT7, ET7) under acidic conditions demonstrated the critical role of the N-terminal ammonium group on the trimeric folds. This finding is consistent with a previous study by Wennemers et al., which showed that the N-terminal ammonium groups significantly destabilize the collagen triple helix.¹³ acKT7 forms the most stable triple helix at pH 11, while acET7 forms the most stable one at pH 3, further demonstrating that a neutral N-terminus is largely favored for a collagen triple helix.

In order to have more insights into the interactions between the charged side chains and the terminal functional groups, we did measure the folding enthalpy change (ΔH) of a triple helix for these CMPs using differential scanning calorimetry. However, the measured ΔH values were not linearly correlated with the T_m values. This could be due to the hydration of the

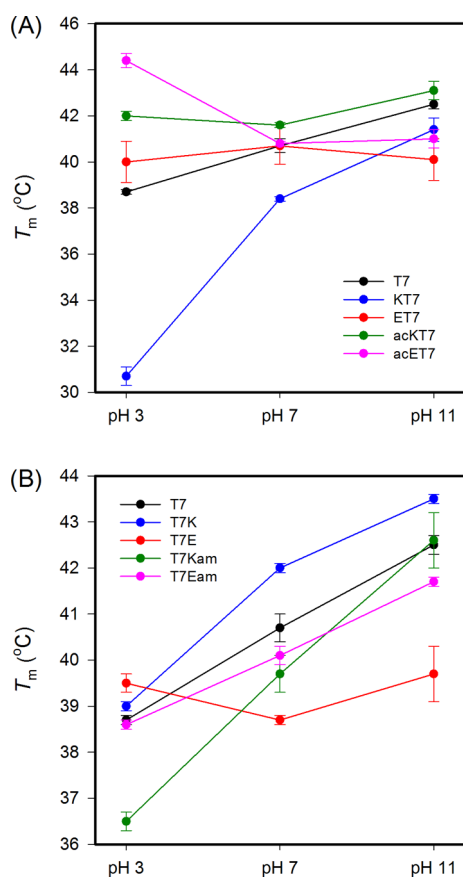


Figure 3. Plots of the T_m value versus pH for (A) KT7, ET7, acKT7, and acET7 and (B) T7K, T7E, T7Kam, and T7Eam. The parent peptide T7 is included in each plot for comparison. The error bars indicate the deviations of T_m values.

charged side chains, which made the ΔH values not correspond with the T_m values. Thereby, we used the T_m values as the index to discuss our results and observations for all the CMPs studied in this work.

Effects of the C-Terminal Charged Residues on the Triple Helix Stability. We also prepared two peptides (T7K and T7E) with a free carboxylic group at the C-terminus and two amidated peptides (T7Kam and T7Eam) to examine how a charged side chain at the C-terminus would affect the triple helix stability. According to the CD spectra and sigmoidal thermal unfolding curves (Figures S1, S2, and S3 of the Supporting Information), all of these peptides folded into a triple helix. As shown in Table 2 and Figure 2D–F, T7K had a T_m value of 42.0 °C at pH 7, which was higher than that of the control peptide T7 (40.7 °C), suggesting that the **C-terminal Lys residue formed a favorable electrostatic interaction with the terminal α -COO[−]** through its side chain γ -NH₃⁺ to stabilize the triple helix. The existence of such an electrostatic interaction was further confirmed by the fact that the T_m value of T7K decreased upon protonating the C-terminal carboxyl group at pH 3 or adding concentrated salts to break down the electrostatic interaction. Further evidence demonstrating the contribution of Lys ammonium–carboxylate interactions was that the amidated peptide T7Kam formed a less stable triple helix than did T7K (39.7 °C vs 42 °C) at neutral pH. Moreover, we also found that the increase in the T_m value for T7Kam was quite significant (~6 °C) between pH 3 and pH 11 (Figure 3B), implying that a positively charged Lys side

chain is unfavorable to the triple helical fold at the C-terminus when the terminal functional group is neutral. This is likely due to the interchain Lys–Lys repulsions decreasing the triple helix stability.

Regarding the peptide with a Glu residue incorporated into the C-terminus, T7E, we found that its T_m value at pH 7 decreased by approximately 2 °C in comparison with that of the control peptide T7, indicating that the side chain carboxylate of Glu caused negative–negative charge repulsions with the terminal carboxylate to destabilize the triple helix. The unfavorable repulsion effects could be diminished in the presence of 2 M NaCl or protonating the carboxylic groups. The pH-dependent measurements showed that T7E did not have significant changes in its triple helix stability, which was different from the findings for KT7. This result suggests that the charge repulsions at the C-terminus imposed a smaller destabilization effect on the triple helix than did those at the N-terminus. In addition, as shown in Figure 3B, T7Eam and T7 had close T_m values under each pH condition and exhibited similar changes in the T_m value from pH 3 to pH 11, suggesting that the influences of a Glu side chain and a C-terminal carboxylate on the triple helix were comparable. Furthermore, T7Kam displayed greater pH-dependence in stability than did T7Eam, in which the ΔT_m between pH 3 and pH 11 were 6.1 °C for T7Kam and 3.1 °C for T7Eam (Figure 3B). This finding demonstrates that a more localized charge of primary amine has a higher impact than the more delocalized charge of the carboxylate group on the triple helix from the C-end. Moreover, as shown in Figure 3, T7E and T7K formed a more stable trimer than did their counterparts (T7Eam, T7Kam) at pH 3, indicating that a neutral carboxyl group (–COOH) at the C-terminal end favors the triple helical conformation. This also concurs with a previous study that demonstrated the positive contribution of a neutral carboxyl group at the C-terminus to collagen folding.¹³

Effects of Side-Chain Length and Combinational Consequences on the Triple Helix Stability. To explore whether the side-chain length would play a critical role in the terminal electrostatic interactions and the triple helix, we prepared another four peptides with Asp or a noncanonical amino acid, ornithine (Orn), incorporated into the termini (OrnT7, DT7, T7Orn, and T7D). Essentially, such a substitution would not change the charges on the peptides because Asp and Glu have similar side chain pK_a values (3.9 vs 4.3),³⁶ while Orn and Lys possess closer side chain pK_a values (10.8 vs 10.4).^{36,37} Because the side chain of Asp is one methylene shorter than that of Glu and Orn has one less methylene than Lys in the side chain, we could examine the length effects on the folding of a collagen triple helix. Based on the CD spectra and thermal denaturation curves (Figures S4 and S5 of Supporting Information), these peptides all formed triple helical assemblies. From the melting temperatures measured at different pH values, all the Lys to Orn substituted CMPs exhibited similar tendencies in pH-dependent stability (Figures 4 and 5). Likewise, the Asp-containing CMPs possessed a pH-dependent stability trend similar to that of the Glu-containing CMPs (Figures 4 and 5). The results indicate that, for a same-charged side chain, its length does not alter the trend in the pH-dependent stability of collagen triple helices.

Although shortening the charged side-chains did not change the pH-dependent properties for the CMPs, the impact on the N-terminus was distinct from that on the C-terminus. As

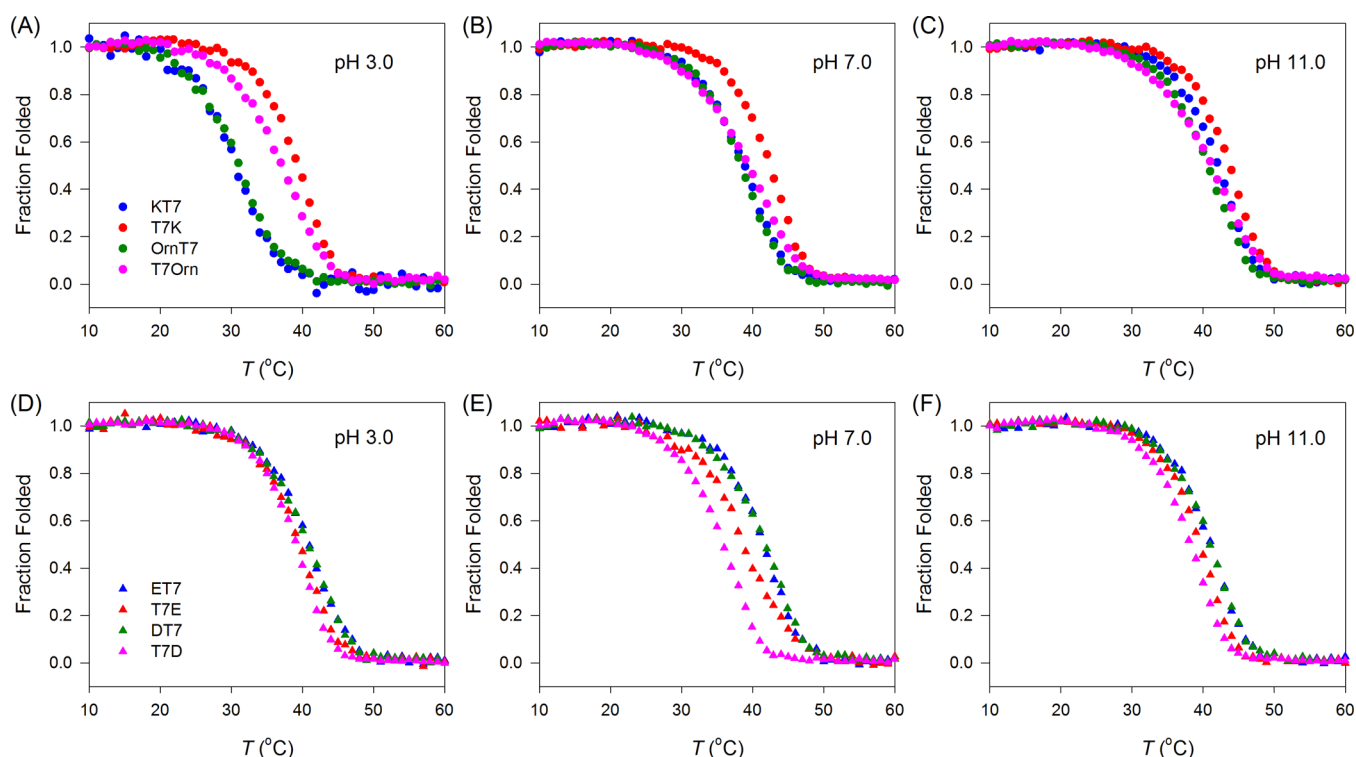


Figure 4. Thermal unfolding curves of the triple helices derived from CMPs bearing a terminal Lys, Orn, Glu, or Asp residue at different pH values. The curves are normalized to the folded fraction.

shown in Figure 4 and Table 2, the difference in T_m values is within 0.3 °C between KT7 and OrnT7 in the positively charged forms (pH 3 and pH 7), while that difference is smaller than 1.0 °C between ET7 and DT7 in negatively charged forms (pH 7 and pH 11), showing that the side-chain length is not a crucial factor affecting the N-terminal charge–charge interactions and the triple helix stability. In contrast, when these charged residues were installed into the C-terminus, the T_m value of T7K was 2–2.6 °C higher than that of T7Orn with a positively charged side chain (pH 3 and pH 7), while the T_m value of T7E was 1.8–2.9 °C greater than that of T7D with a negatively charged side chain (pH 7 and pH 11). These results suggest that the side-chain length could be an important element to modulate the C-terminal charge–charge interactions and the triple helix stability. At neutral pH, Lys and Orn have a protonated side chain which may form electrostatic interactions with the C-terminal α -COO[−] to favor the triple helical fold. The experimental data showed that T7K formed slightly more stable triple helices than the control peptide T7, while T7Orn formed less stable triple helices than T7, suggesting that the longer side chains of Lys might have been closer to the terminal carboxylate and generated a more favorable electrostatic interaction than the side chains of Orn. At acidic pH, the C-terminal carboxyl group is protonated and neutral, but Lys and Orn carry a positively charged side chain. The shorter side chain of Orn may induce more intense interchain charge–charge repulsions during the folding of a triple helix, thus making T7Orn less stable than T7K at pH 3. At neutral and alkaline pH, both Glu and Asp have a deprotonated side chain that would generate charge–charge repulsions with the C-terminal carboxylate and with the negatively charged side chains on the adjacent strands upon the formation of a triple helix. Compared to the control peptide T7, both T7E and T7D formed less stable triple

helices, indicative of a destabilizing side chain–C-terminus interaction, and the triple helices derived from T7D were less stable than those derived from T7E. Because Asp has a shorter side chain than Glu, the side chains of Asp residues on adjacent strands within a triple helix were most likely close to each other and thus induced stronger repulsions to make T7D less stable than T7E.

Furthermore, we prepared a peptide ET7K that contained an anionic residue at the N-terminus and a cationic residue at the C-terminus to investigate if the stabilization effect was additive. Meanwhile, we also synthesized another peptide KT7E with a N-terminal cationic residue and a C-terminal anionic residue, installed to examine whether the destabilization effect was additive. The CD spectra and thermal unfolding curves (Figures S6 and S7) showed that the two peptides formed triple helices. As shown in Figure 6 and Table 2, ET7K had a T_m value of 41.6 °C at pH 7, which was only slightly higher than those of T7 and ET7 but lower than that of T7K, showing that there was no significant additive stabilization effect. In contrast, the T_m value of KT7E was 34.2 °C at pH 7, which was lower than those of T7, KT7, and T7E, indicating that the destabilization effect was additive.

CONCLUSIONS

In this work, we designed and synthesized a series of CMPs to perform a comprehensive study on the effects of terminal charged residues on a collagen triple helix. The results showed that favorable electrostatic interactions can be induced between the side chain ammonium of a C-terminal Lys residue and the terminal carboxylate to stabilize the triple helix. In contrast, such electrostatic interactions between the side chain carboxylate of an N-terminal Glu residue and the terminal ammonium are relatively weak, so they do not confer notable stabilization effects. In addition, the positive charges

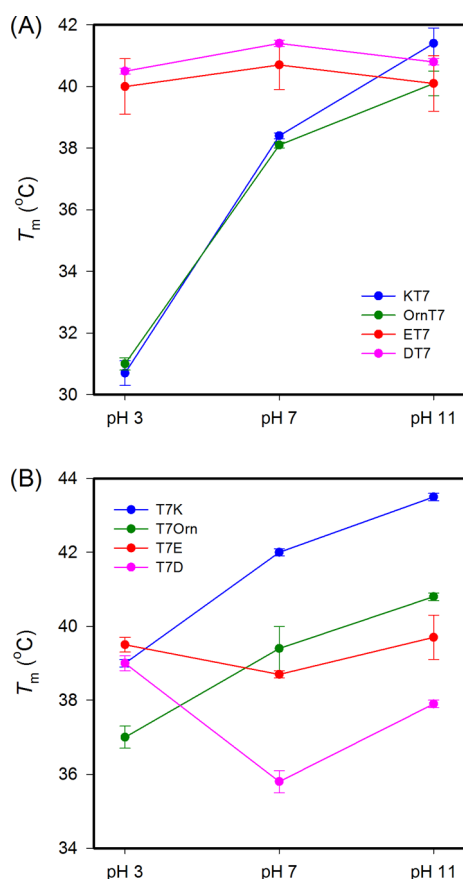


Figure 5. Plots of the T_m value versus pH for (A) KT7, OrnT7, ET7, and DT7 and (B) T7K, T7Orn, T7E, and T7D. The parent peptide T7 is included in each plot for comparison. The error bars indicate the deviations of T_m values.

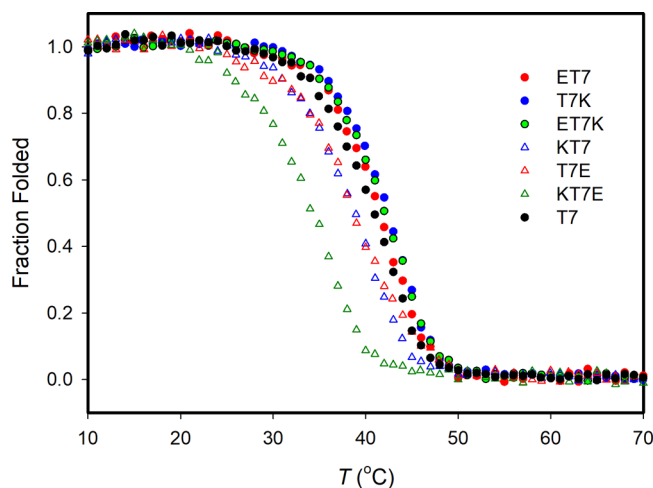


Figure 6. Thermal unfolding curves of the triple helices derived from CMPs bearing a Lys or Glu residue at one terminus and combining both Lys and Glu residues in a peptide at pH 7. The curves are normalized to the folded fraction and the reference peptide T7 is included for comparison.

between the side chain ammonium of Lys and the N-terminal ammonium produce unfavorable electrostatic repulsions to destabilize the triple helix, while the negative charge repulsions between the side chain carboxylate of Glu and the C-terminal carboxylate impose a destabilization effect on the trimeric fold.

Although either ammonium–ammonium or carboxylate–carboxylate repulsion leads to a less stable triple helix, the impact of N-terminal ammonium–ammonium repulsion seems more pronounced. As for the effect of the side-chain length, our data showed that shortening a methylene from glutamate to aspartate produced a greater impact than did the shortening from lysine to ornithine, demonstrating that replacing glutamate with aspartate significantly varied the distance between the side chain $-\text{COOH}$ and the terminal $-\text{NH}_2$ or $-\text{COOH}$. Furthermore, the stabilization effect of terminal electrostatic interactions is not additive, but the destabilization effect of terminal charge repulsions is. **In conclusion, we have shown that the stability of a collagen triple helix can be modulated by installing a cationic or anionic residue into the terminus of a peptide,** which provides valuable information for the design of collagen-related peptides and materials.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acs.jpcb.1c01631>.

Molecular weights of the peptides, the raw (not normalized) CD-monitored thermal denaturation curves, and the fitting method for the determination of T_m values (PDF)

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Notes

The authors declare no competing financial interest.

■ ACKNOWLEDGMENTS

We are grateful to the financial support from the Ministry of Science and Technology of Taiwan (MOST 107-2113-M-007-002 & MOST 108-2113-M-007-012). This work was also financially supported by the “Frontier Research Center on Fundamental and Applied Sciences of Matters” from the Featured Areas Research Center Program within the framework of the Higher Education Sprout Project (110QR001IIS) by the Ministry of Education (MOE) in Taiwan.

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